Session objective : Overview of GenAI

- ☑ Business problem
- ☑ Rule based solution
- ☑ mathematical Approach
- ☑ ML
- ☑ Deep learning
- ☐ LLM
- ☑ Next ? PE

70 yrs

XYZ
↓
data
↓

C1 → loan

income, cibil-score → Domain → Approve / Reject

100%

in a day → 10 customers

C1 → (_ , _) → Ap / Rejct

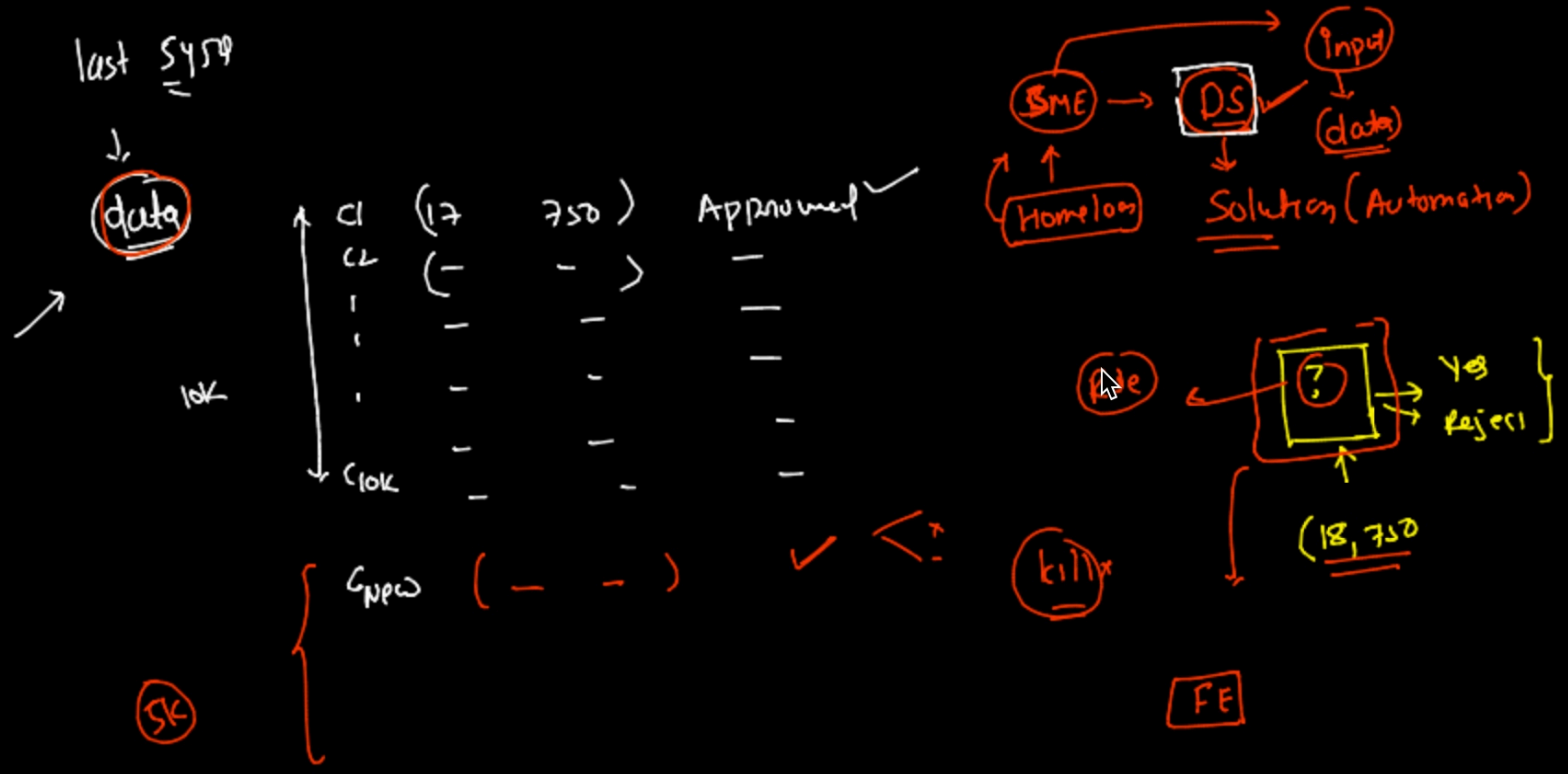
C10 ( )

100 … 1000

C1 ( )

C1000 ( )

time


last 5yrs
data
10K
5K
C1 (17 750) Approved
C2
C10K
Cnew
SME
DS
input
data
Homeloan
Solution (Automation)
Yes
Reject
(18, 750
FE

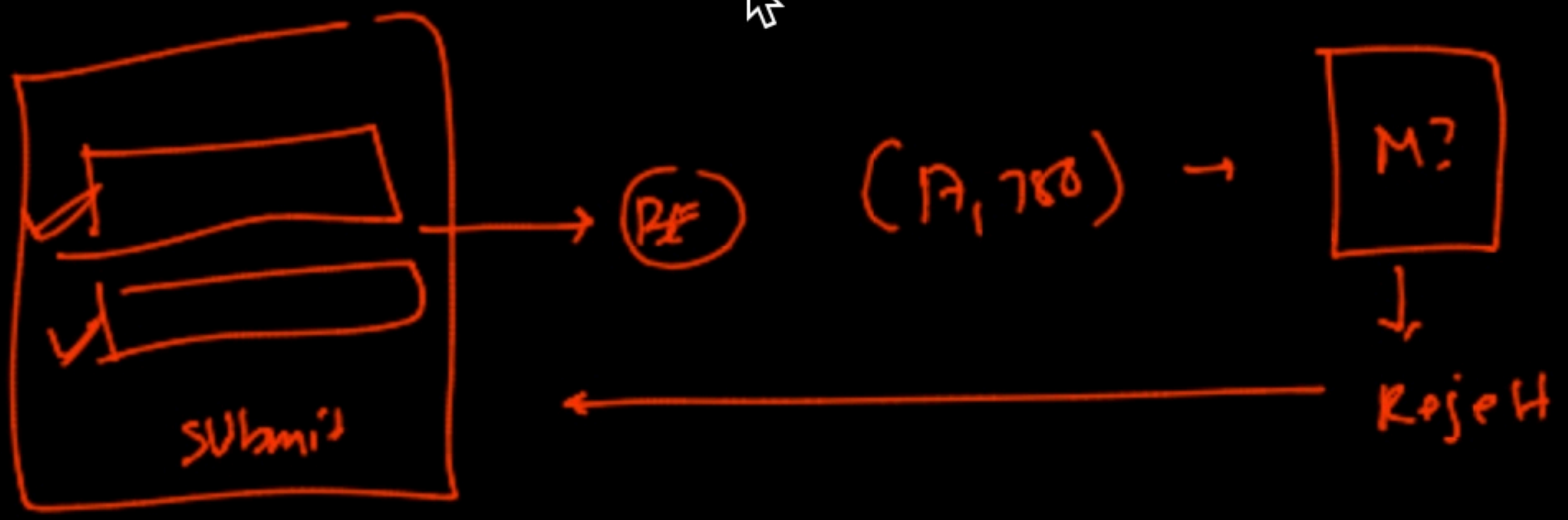
HDFC Bank

Portal

↓

Apply loan

↓



1000

20k

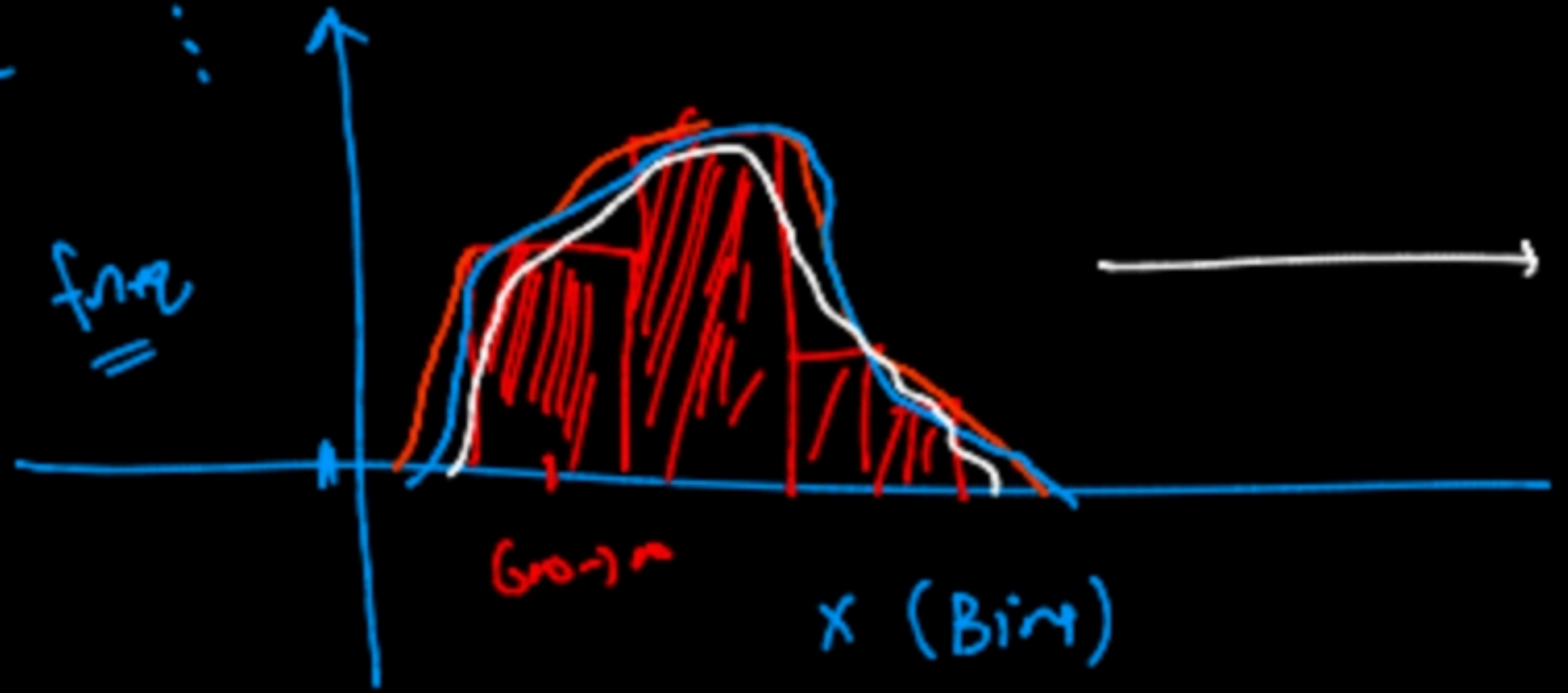
Histogram
↑
X (cibil-score)
↓
610
⋮

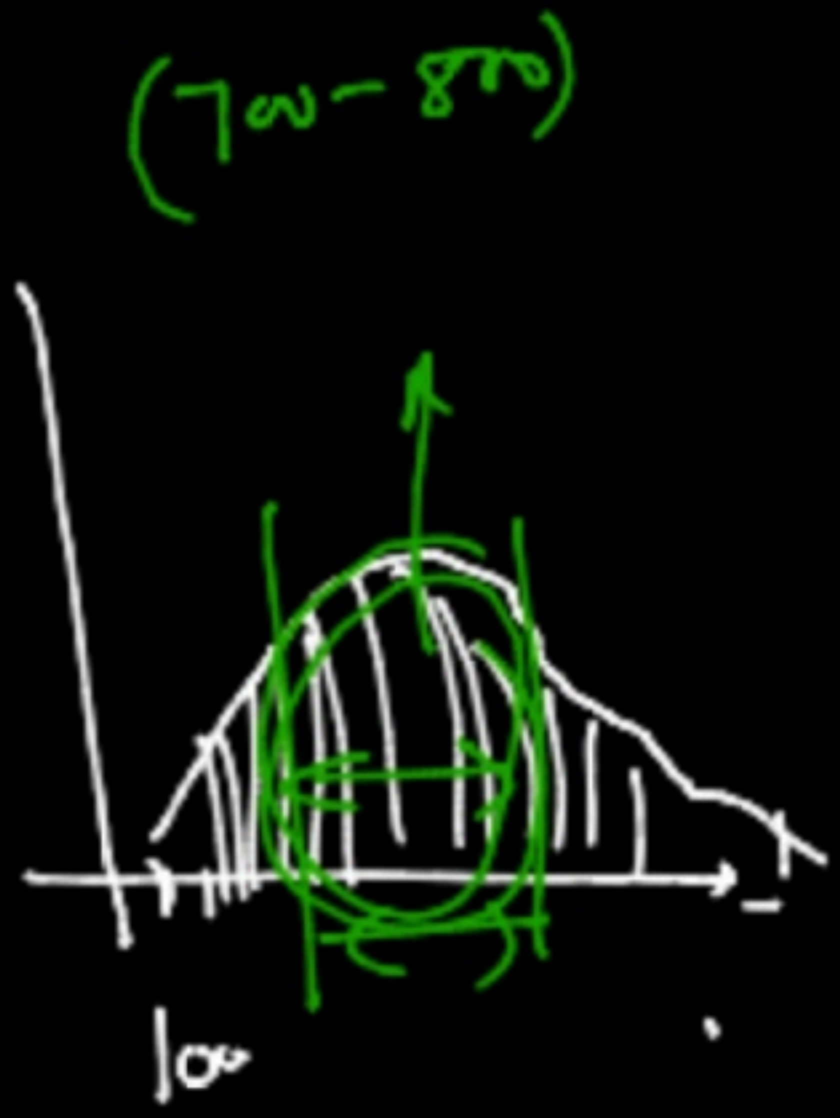
50

→ Bin (5-10)

Bin

600 - 700 → 24
700 - 800 → 24
⋮





code

Matplotlib
or
Seaborn

KDE → Kernel Density estimation

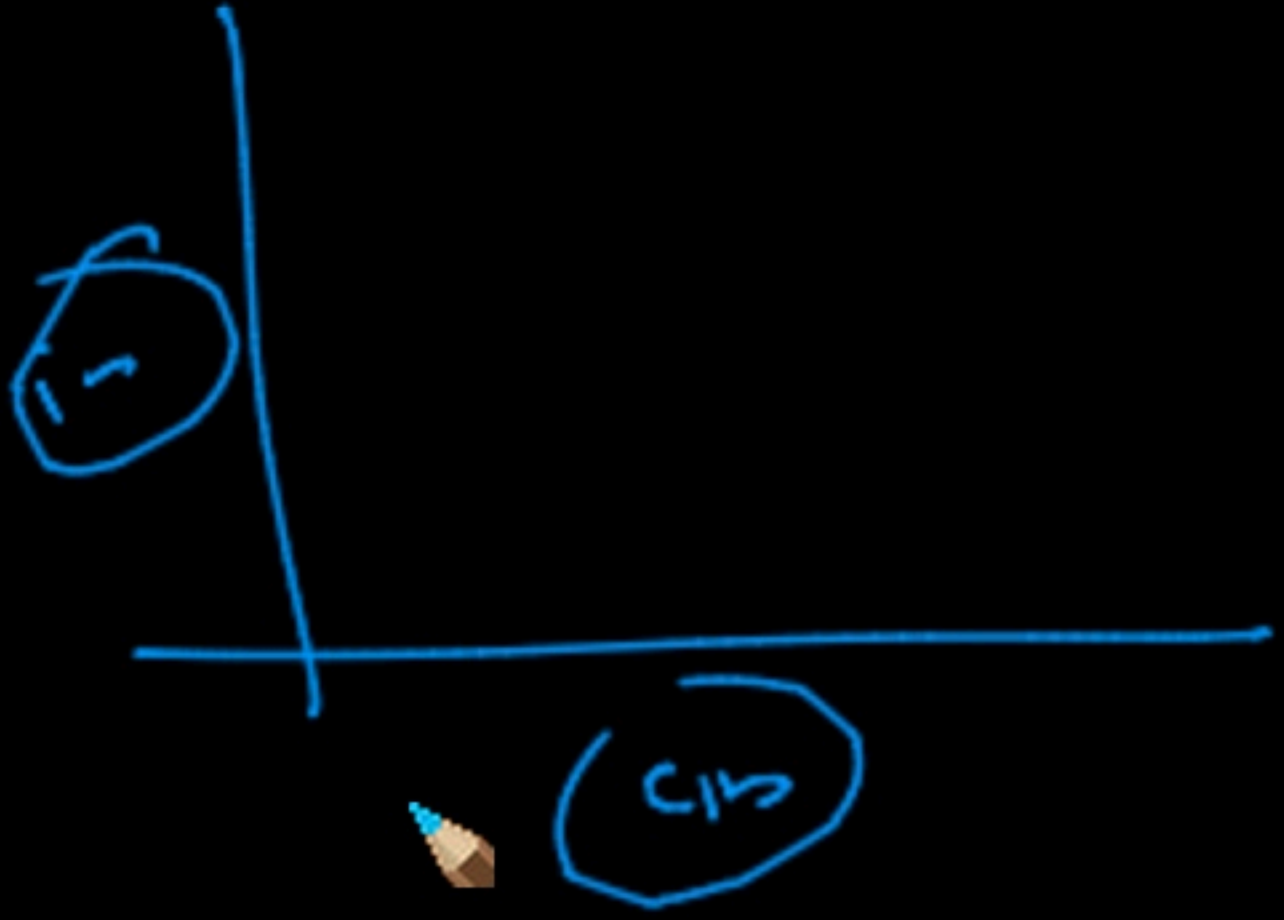
1. cibil-score → conclusion (Rule) →
2. Income → Rule
3. # of's → Rule
4. → Rule

math →

If ——

If ——

1. Univariate Analysis
2. Bivariate Analysis



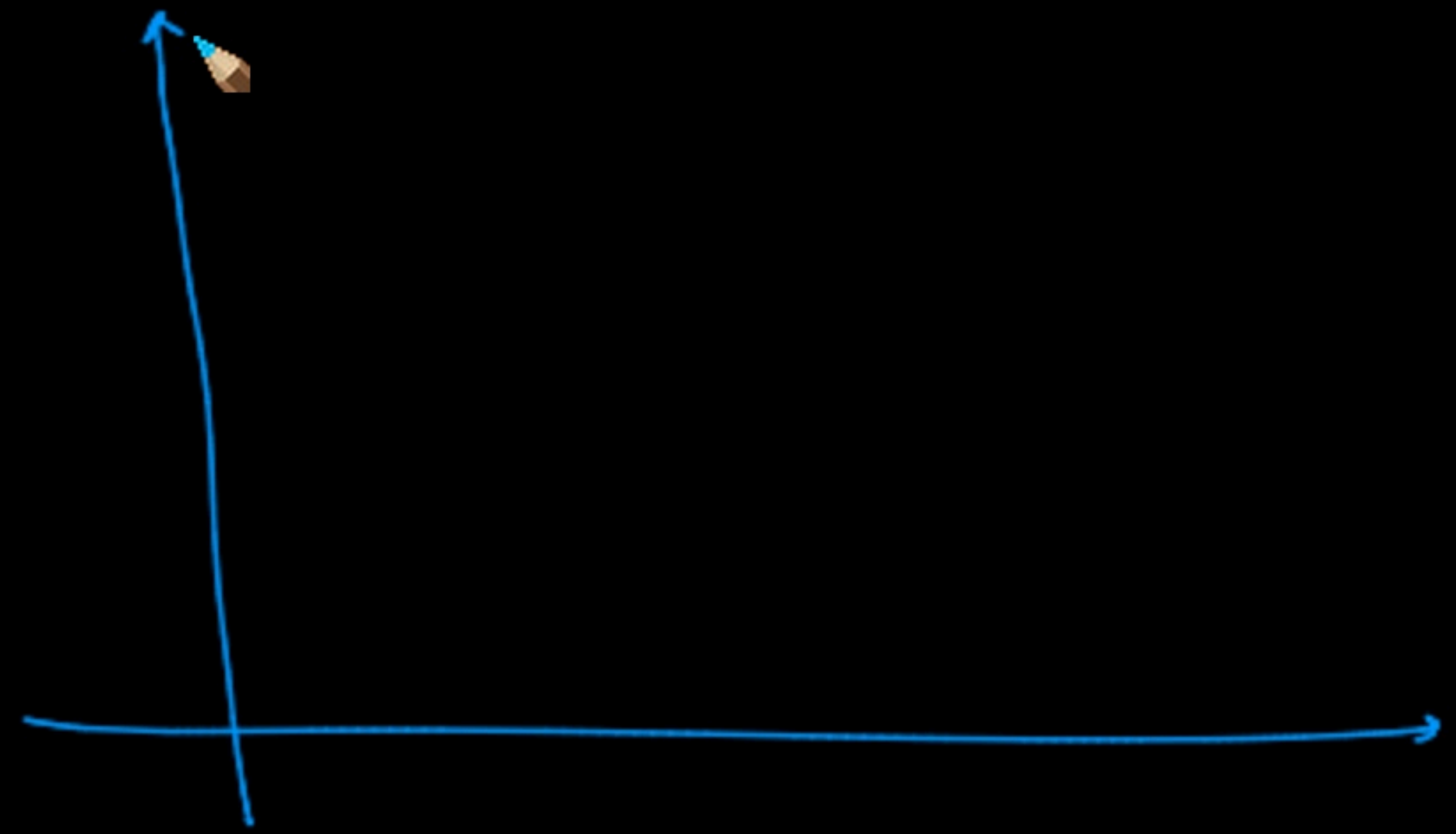
ML ×

=

≈

Math

eg. →



	a_1	a_2		
5 C	17	650	Yes	$\sqrt{(17-7.8)^2 + (498-650)^2} \Rightarrow 7.8$
6 C	16	750	Yes	8.8
4 C	15	780	Yes	5.6
3 C	3	410	No	0.98
1 C	4	590	No	0.23
2 C	3	480	No	0.96
7 C_7	22	698	Yes	7.9

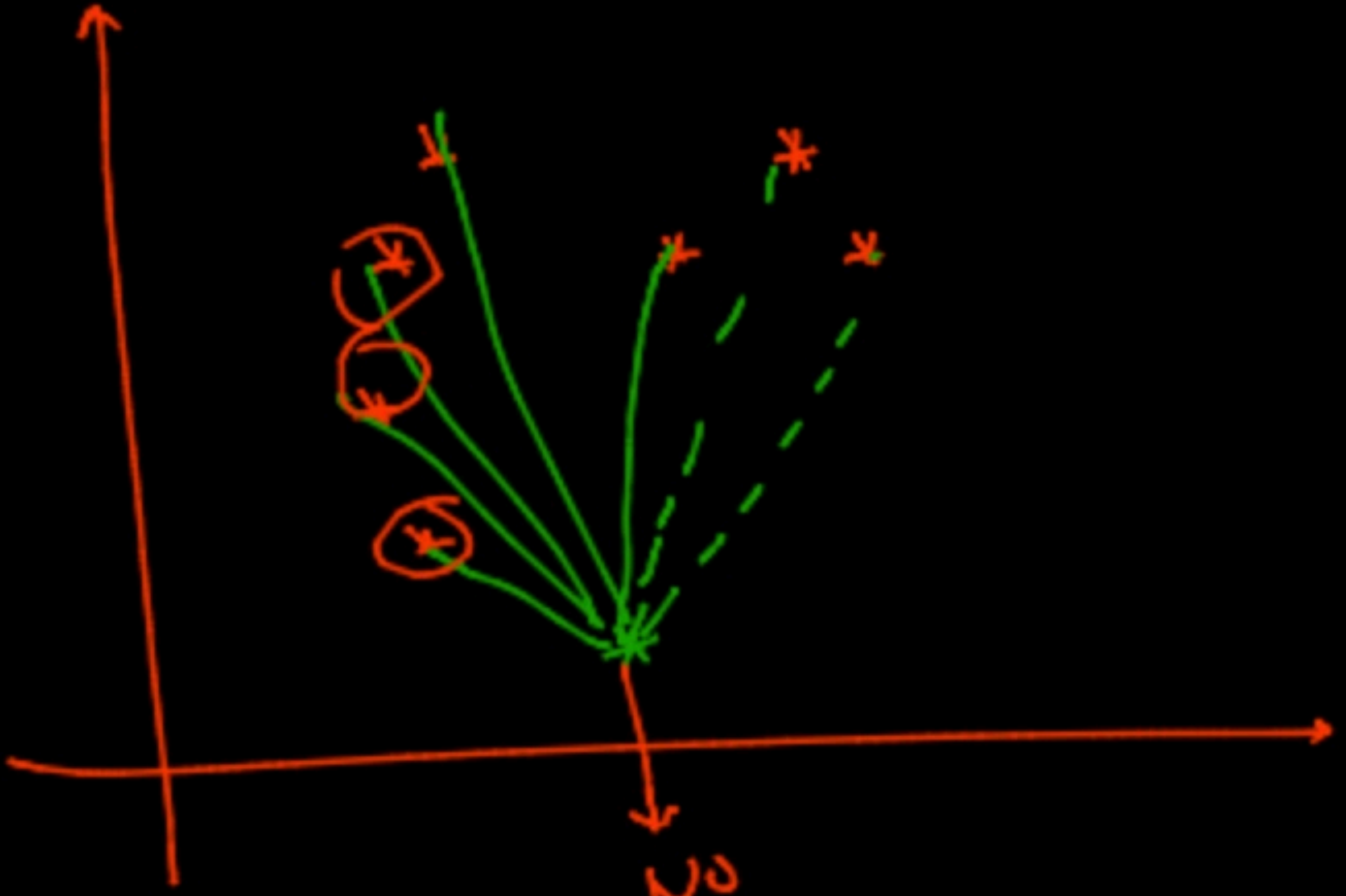
$C_8 \rightarrow [7.8, 498] \rightarrow$ No

No → ✓
Yes → ✓

No → 1
Yes → 0

1!

100%



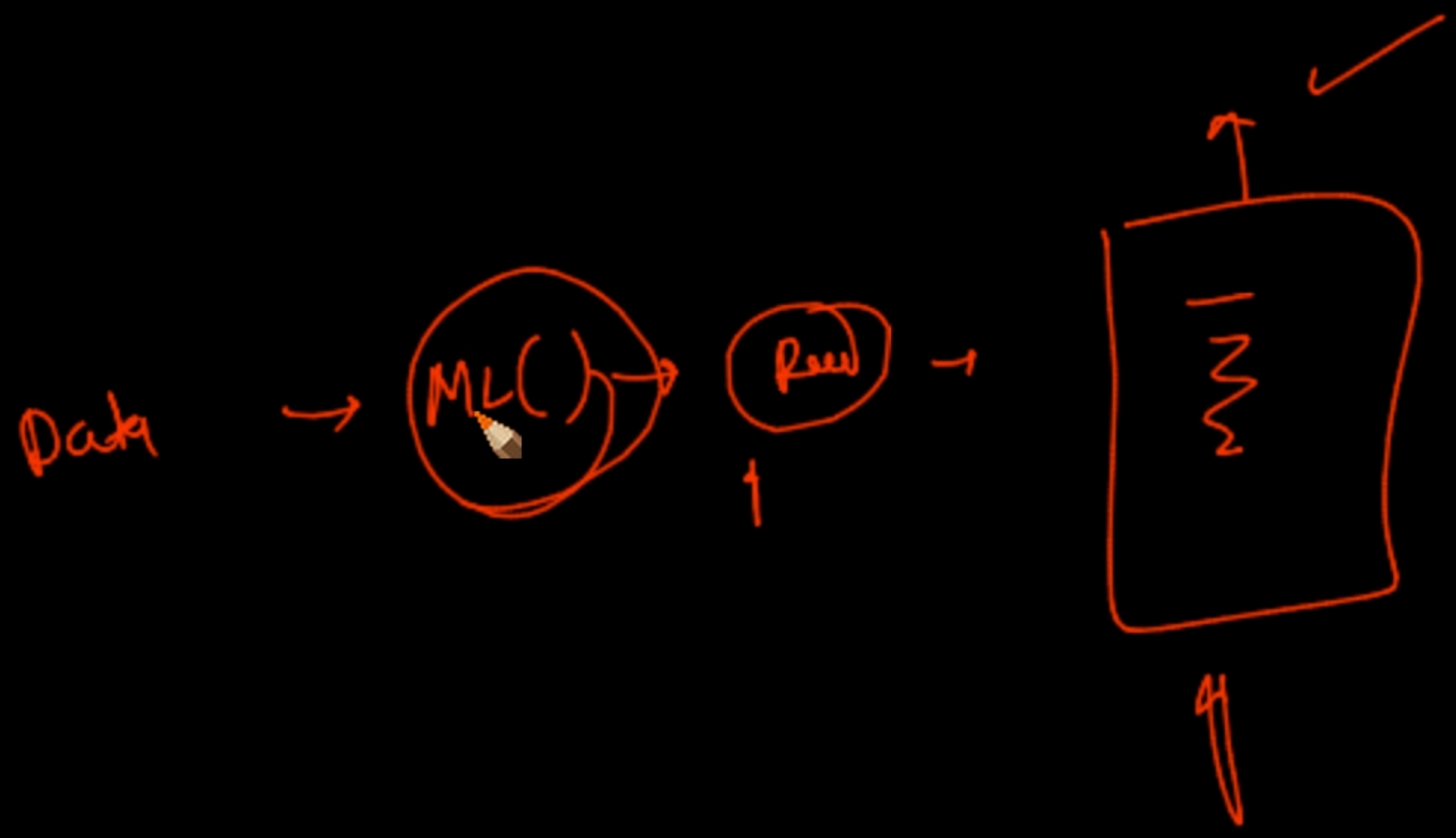
No

Rule → math (distance)

distance b/w 2 points

~~p(a₁a₂)~~ $p(a_1, a_2)$, $q(b_1, b_2)$

$$d(p,q) = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2}$$ ✓


Data
ML( )